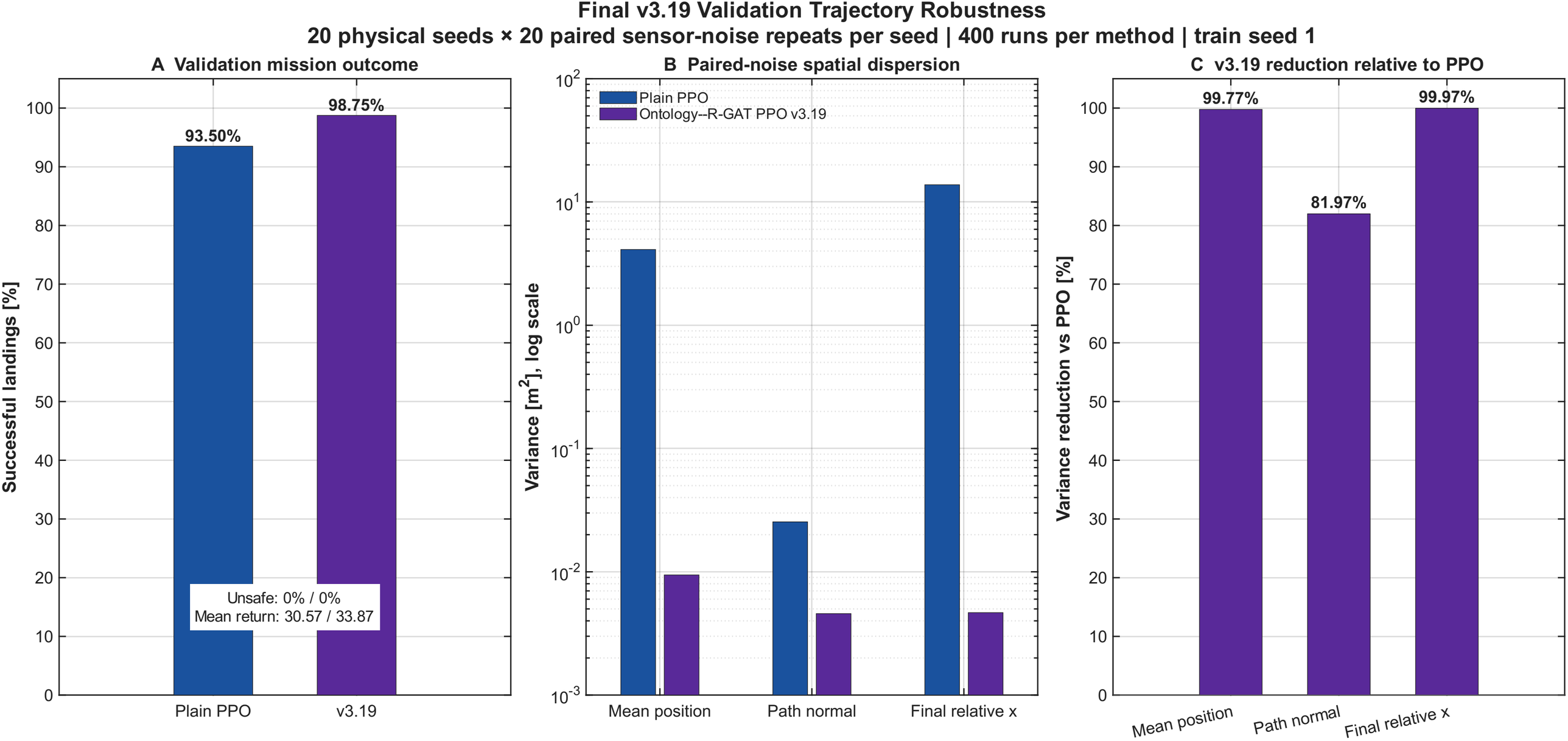

Final v3.19 Validation Trajectory Robustness
20 physical seeds × 20 paired sensor-noise repeats per seed | 400 runs per method | train seed 1
A Validation mission outcome
Successful landings [%]
93.50%
98.75%
Unsafe: 0% / 0%
Mean return: 30.57 / 33.87
Plain PPO
v3.19
B Paired-noise spatial dispersion
Variance [m^2], log scale
Plain PPO
Ontology--R-GAT PPO v3.19
Mean position
Path normal
Final relative x
C v3.19 reduction relative to PPO
Variance reduction vs PPO [%]
99.77%
81.97%
99.97%
Mean position
Path normal
Final relative x